UNCERTAINTY QUANTIFICATION OF AN AEROELASTIC MODEL WITH SPARSE POLYNOMIAL SURROGATES

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In this paper we study aircraft aeroelastic interactions and the propagation of parametric uncertainties in numerical simulations using high-fidelity fluid flow solvers. We more particularly address the influence of variable operational and structural parameters (random inputs) on the drag performance and shape (outputs) of a flexible wing in transonic regime. Polynomial surrogate models based on homogeneous chaos expansions in the random inputs are considered in this respect. The polynomial expansion coefficients are usually constructed by projection using either structured sampling sets of the input parameters, as Gauss quadrature nodes, or unstructured sampling sets, as in Monte-Carlo methods. However, in complex systems such as the advanced aeroelastic test case studied here, the output quantities of interest generally depend only weekly on the multiple cross-interactions between the random inputs. Consequently, only low-order polynomials significantly contribute to their surrogates, which thus have a sparse structure in the underlying polynomial bases. This feature prompts the use of compressed sensing for the construction of the polynomial surrogates by regression. This alternative methodology is non-adapted and considers unstructured sampling sets orders of magnitude smaller than the structured or unstructured sampling sets required in projection methods. It is illustrated in the present work for a moderately to high dimensional parametric space and an aeroelastic test case of industrial relevance.

Keywords: Computational fluid dynamics, aeroelasticity, polynomial chaos, compressed sensing.

1. Introduction

Aeroelasticity deals with the coupling of a flexible structure and a fluid flow and involves complex, possibly highly non-linear physical phenomena. The assessment of these non-linear effects is of crucial importance for the stability and strength of the structure, such as aircraft, bridge, turbine, *etc*., since they may lead to its complete destruction in case of self-sustained (limite-cycle) or divergent (flutter) oscillations. Otherwise, fluid flows induce structural deformations which in turn influence the fluid dynamics such as for example shock behavior at transonic speeds, which in turn impacts the performance of the profile. Aeroelastic interactions have thus to be accounted for in design, testing, certification, and subsequent optimization if need be. However several sources of variability arise in the prediction of aeroelastic phenomena, including measurement uncertainties during ground or flight testing, parametric uncertainties, or model and numerical uncertainties pertaining to analysis methods. Uncertainty quantification (UQ) and the various techniques it encompasses are aimed at estimating quantitatively all

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those sources of uncertainties or variabilities and predicting their significance on quantities of interest used to design, test, certify or optimize the structure. UQ is generally addressed in a probabilistic framework, although the actual regulations do not rely on probabilities for certification but rather introduce the notion of risks related to undesirable (failure) scenarios. Recent reviews on the various issues raised by UQ in aeroelasticity are given in Badcock *et al*. (2011), Beran *et al*. (2017), Dai & Yang (2014), Pettit (2004). Applications to stability or optimization are described in Arnaud & Poirion (2014), Castravete & Ibrahim (2008), Hosder *et al*. (2008), Kuttenkeuler & Ringertz (1998), Le Meitour *et al*. (2012), Poirion (1992), Witteveen *et al*. (2007) for example. In this paper the influence of operational and structural parametric uncertainties (inputs) on the shape and aerodynamic performances (outputs) of a flexible wing in transonic regime, is quantified. This UQ analysis is performed thanks to polynomial surrogate models of the quantities of interest accounting for fluid-structure interactions. The air flow is solved by high-fidelity computational fluid dynamics (CFD) tools while the wing motion and stresses are computed by a simplified beam model. The coupling is ensured by balancing the aerodynamic loads for a prescribed lift force, considered as a variable input parameter, with the efforts in the wing for its induced shape. The overall procedure is outlined in Ghazlane (2012), Marcelet (2008).

Polynomial surrogate of random functionals are typically obtained invoking the homogeneous, or polynomial chaos (PC) expansion introduced by Wiener (1938) for stochastic processes. It is defined as the span of Hermite polynomial functionals of a Gaussian random variable. Mean-square convergence is guaranteed by the Cameron-Martin theorem (Cameron & Martin, 1947). It is optimal (*i.e.* exponential) for Gaussian processes but sub-optimal for non-Gaussian processes, as shown numerically in Xiu & Karniadakis (2002). Generalized polynomial chaos (gPC) expansions with other families of polynomials which numerically recover the foregoing optimality property have then been studied in Ernst *et al*. (2012), Soize & Ghanem (2004), Xiu & Karniadakis (2002). They consist in expanding a function of random variables into a linear combination of orthogonal polynomials with respect to the probability density functions (PDFs) of these underlying random variables. PC and gPC expansions are now widely used in engineering sciences as a constructive tool for representing random vectors, matrices, tensors or fields, and quantifying uncertainties in complex systems; see Clouteau *et al*. (2001), Ghanem (1999), Ghanem & Spanos (1991), Ghiocel & Ghanem (2002), Le Maître and Knio (2010), Najm (2009), Sun (1979) and references therein. Applications to aeroelastic stability are considered in Hosder *et al*. (2008), Le Meitour *et al*. (2012), Witteveen *et al*. (2007) for example.

Non-intrusive UQ techniques are typically considered in CFD, because the complex flow solvers are preferably treated as black boxes in order to compute the aerodynamic loads and other quantities of interest. Regarding non-intrusive PC and gPC expansions, two approaches for computing the polynomial coefficients of the output quantities of interest have usually been considered: (i) the projection approach, in which they are computed by structured quadratures, *i.e.* Gauss quadratures, or unstructured quadratures, *i.e.* Monte-Carlo or quasi Monte-Carlo sampling; and (ii) the regression approach, minimizing some error measure or truncation tolerance of the polynomial expansion for some particular values of the inputs (which can be the quadrature sets invoked just above, for example). Both techniques suffer from the so-called "curse of dimensionality" when the dimension D of the parameter space increases. Indeed, a polynomial expansion of total degree q in D variable parameters contains $P+1 = \binom{q+D}{q} \approx \frac{D^q}{q!}$ coefficients for D large. A direct way to compute them is to use a tensor product grid in the parameter space requiring about $N \approx \lfloor \frac{q}{2} \rfloor^D$ evaluations of the whole model, where $\lfloor \cdot \rfloor$ stands for the floor function. These N runs are very often unaffordable for large parameter spaces and complex configurations, as in the present case dealing with three-dimensional, non-linear fluid-structure interactions. Smolyak's algorithm (Smolyak, 1963) introduces sparse grid quadratures involving $N \approx \lfloor \frac{q}{2} \rceil^{\log D}$ points while preserving a satisfactory level of accuracy. In Savin *et al*. (2016) it has been observed that such sparse rules typically become competitive with respect to tensor grids for dimensions $D \geq 4$. Consequently, collocation techniques with sparse quadratures or adaptive regression strategies have been developed in order to circumvent the dimensionality concern (Blatman & Sudret, 2010, Ghiocel & Ghanem, 2002, Xiu & Hesthaven, 2005).

In this article we adopt the regression approach to propagate uncertainty in the aeroelastic computations. We also aim at benefiting from the sparsity of the processed outputs themselves (wing shape, drag and pitching moment coefficients) to reconstruct their gPC representations in a non-adaptive way (Doostan & Owhadi, 2011). Indeed, we rely on the common observation that many cross-interactions between the input parameters are actually smoothened, or even negligible, once that have been propagated to some global quantities of interest processed from complex aerodynamic computations. The corresponding polynomial expansions should thus involve only low-order polynomials, such that the contribution of the higher-order polynomials is negligible. We can therefore expect to achieve a successful output recovery by the techniques known under the terminology of compressed sensing (Candès *et al.*, 2006, Donoho, 2006). In this theory the reconstruction of a sparse signal on a given, known basis requires only a limited number of evaluations at randomly selected points–at least significantly less than the dimension P of the basis. We thus resort to unstructured sampling sets to recover sparse outputs. Compressed sensing is formulated as a constrained, underdetermined system which may be solved by standard optimization algorithms.

The rest of the paper is organized as follows. In Sect. 2 the aeroelastic database for which surrogates are sought for is outlined. The aerodynamic model, the structural model, and the computational procedure for flow-structure interactions are introduced. The variable input parameters for this problem, and the sampling strategy for the computations are also detailed. Then Sect. 3 addresses the construction of polynomial surrogates for the output quantities of interest identified in the foregoing section. Probability density functions and sensitivity indices are typically computed using those polynomials surrogates. The application of these techniques of uncertainty quantification to the aeroelastic database considered in this work is presented in Sect. 4. Some conclusions are finally drawn in Sect. 5.

2. Aeroelastic database

In this section the numerical procedure and tools considered to generate the aeroelastic database to be used as a basis for uncertainty propagation, are outlined. The database consists of high-fidelity CFD-elastic simulations representative of an aircraft wing. A sampling of the aeroelastic input parameters (denoted by ξ throughout) covers a very large design space that enables to achieve significant aerodynamic output at cruise (the main output being the drag sensitivity to aerodynamic and structural parameters).

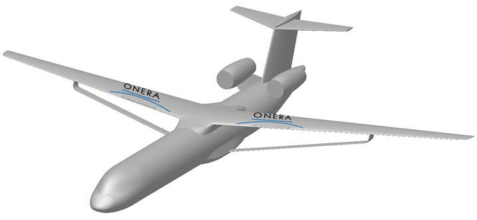

Fig. 1. ALBATROS configuration.

2.1. *The ALBATROS wing*

The configuration of interest is the ALBATROS configuration; see Fig. 1 (Carrier *et al.*, 2012). It has been designed at ONERA in an internal project between 2010 and 2013. The ALBATROS acronym can be translated into "laminar strut braced wing with reduced drag by multidisciplinary optimization" (Aile Laminaire hauBAnée à Traînée Réduite par Optimisation multidiSciplinaire). The purpose of this project was to design and investigate the pros and cons of the strut braced wing configuration, with high-fidelity analyses in both the structural response and the aerodynamic loads. The aircraft has been designed for medium range (3,000 Nm) with 180 passengers (equivalent to an Airbus A321). To achieve natural laminar flow on the wing, the wing sweep is limited to 16$^{\mathrm{o}}$ (at 25% of the chord) and thus the design cruise Mach number is $\underline{M} = 0.75$. The design lift coefficient had been set to $\underline{C}_L = 0.60$. These nominal parameters are underlined to stress the difference with their actual, yet unknown values M and C_L; see the subsequent section.

2.1.1. *Aerodynamic model*

For the purpose of the present study, the configuration of Fig. 1 has been simplified. The strut has been removed so that only the wing and fuselage are accounted for; see Fig. 2. Another simplification has been to consider only turbulent flow instead of laminar flow. This choice was made to simplify the simulations because the laminar/turbulent transition was computed using transition criteria in the CFD solver *elsA* (Cambier *et al.*, 2013), and noise would have been added to the database. The operational aerodynamic parameters considered as uncertain for the database are the Mach number $M \in [0.74, 0.76]$ and the lift coefficient $C_L \in [0.55, 0.65]$. These variations are kept within an acceptable range for a cruise.

The aerodynamic mesh is structured and fully coincident. It consists of 9,008,512 cells (9,879,020 nodes). For parallelism purposes, the mesh is decomposed into 240 blocks distributed over 60 processors. The main features of the *elsA* (Cambier *et al.*, 2013) aerodynamic simulations are the following: Spalart-Allmaras turbulence closure model (Spalart & Allmaras, 1992); Jameson centered scheme with additional artificial viscosity ($\chi_2 = 0.5$, $\chi_4 = 0.016$) outside of the boundary layers (Jameson, 1983); backward Euler time integration scheme; implicit LU-SSOR phase (Yoon & Jameson, 1987); and isotropic multigrid acceleration technique (total of three grid levels, each time keeping one node over two). The Courant-Friedrichs-Lewy (CFL) stability condition CFL $= 50$ is chosen for the backward Euler time integration scheme.

The aerodynamic simulations are post-processed to obtain a detailed drag breakdown through the far-field drag technique (ONERA's `ffd01` software, see the theory in Destarac & van der Vooren (2004)). This post-processing allows first to convert the pressure/friction drag breakdown given by the CFD solvers into a physical breakdown (lift induced drag $C_{D,\text{lift induced}}$, viscous drag $C_{D,\text{viscous pressure}} + C_{D,\text{friction}}$ and wave drag $C_{D,\text{wave}}$). The cells containing wave or viscous drag are isolated to allow for specific volume integration of drag; see Fig. 3. This analysis also enables to remove a numerical spurious contribution ($C_{D,\text{spurious}}$) that stems from the imperfect numerical resolution of the gradients (at leading edges, for instance). It is important to isolate this contribution because the mesh is deformed to represent various wing shapes, and thus this contribution evolves from one shape to another. Using this analysis increases the precision of the aeroelastic database; see Hantrais-Gervois & Destarac (2015). The output of the database contains the drag breakdown for each parameter set.

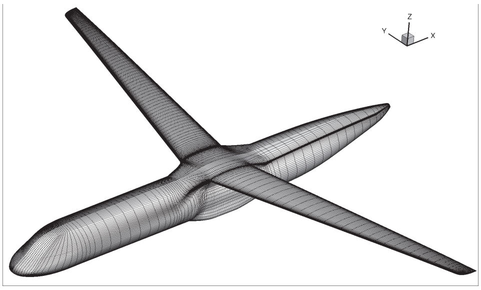

Fig. 2. Aerodynamic surface mesh (mix of the full grid and the one-node-over-two grid).

2.1.2. *Structure model*

As the configuration does not include the strut, no representative structure model exists. Thus, a simplified model has been created for the purpose of this study. Thanks to the work done in Ghazlane (2012) on aero-structure optimization, an equivalent beam model has been derived

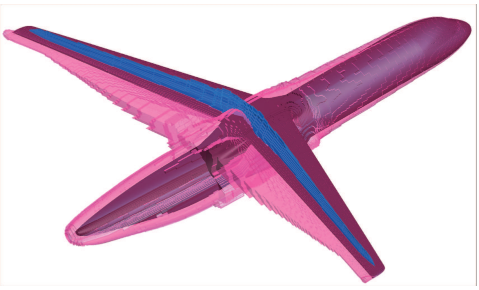

Fig. 3. Far-field drag analysis (Destarac & van der Vooren, 2004). The wave drag volume is in blue and the viscous pressure drag volume is in pink.

(`InAirSsi` module). The nodes of the beam are both displacement and force nodes. They have been determined on the wing only, thus a small offset can be seen between the wing on the configuration and the structure nodes. This deviation is deemed unimportant for the present purpose. A jig shape has been derived from the 1g loading and the beam structure; see Fig. 4. The jig shape is kept fixed for all the different shapes investigated in this study. The meaning of this is that the shape to manufacture is designed once and for all and any structure uncertainty will be visible on the flight shape, which is a relevant scheme for an aircraft process. If the jig shape had been adapted for each structure parameter, the uncertainty on cruise deformation and drag would only come from the jig determination process and would be hardly noticeable. The aeroelastic simulations are carried out from the 1g shape to accelerate the convergence (accounting from the jig to 1g efforts and moments).

The structure design parameters are directly acting on the torsion and bending stiffnesses. Four fixed locations (span-wise) are chosen and the design parameters consist of coefficients to be applied on the stiffness curve; see Fig. 5. Large deviations can be achieved on the flight shape for different stiffness param-

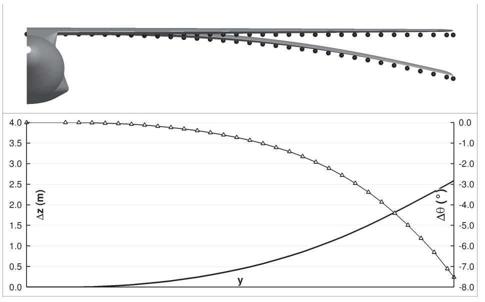

Fig. 4. Structure modes of the beam and jig to 1g loads: deflection (solid line) and torsion (△ line).

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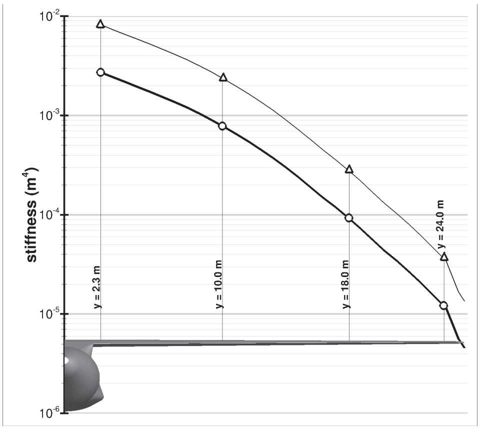

Fig. 5. Bending (○) and polar (△) moments of inertia of the wing in the reference configuration. The vertical lines indicate the location of the control points/cross-sections

2.2. *Aeroelastic computation process*

For each set of input parameters, including the lift coefficient, the wing shape must be at equilibrium for the corresponding lift force. The balancing process is sketched on Fig. 6. Two loops are nested. The outer loop concerns the prescribed lift: the angle of attack α is varied to reach the prescribed lift coefficient. The inner loop concerns the aeroelastic computation itself: for a given angle of attack α, the wing shape corresponding to the equilibrium of the aerodynamic loading and the structural response is sought for through an iterative process. For each iteration, 300 CFD cycles are computed. The integrated efforts and moments are transferred to the beam according to the methodology outlined in Marcelet (2008), and the corresponding beam displacements are computed using a dedicated module of the *elsA* software (Cambier *et al.*, 2013, Marcelet, 2008). The induced aerodynamic mesh deformations are applied thanks to another dedicated analytical tool (`SeAnDef`) described in (Méheut *et al.*, 2012). When convergence is reached, the solution is analyzed with the far-field drag analysis `ffd01` software. The whole process is driven by a `python` script which requires about six hours (wall-clock time) on 60 cores for each set of parameters.

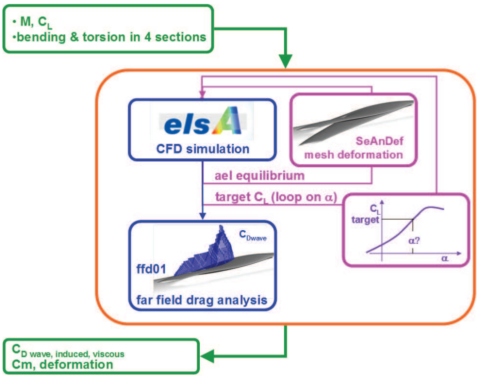

Fig. 6. Aeroelastic process for each set of parameters.

2.3. *ALBATROS database*

2.3.1. *Manual range quantification*

Prior to the high-fidelity computations, the parameter variation amplitudes have been checked through 19 simulations (only one variation at a time). The sensitivity of wing deformation and thus of drag is observed to be far more pronounced for the bending than for the torsion. Therefore larger multiplication factors are applied to torsion to detect effects on the outputs (drag). However, when reaching too high or too low values, the aeroelastic simulations diverge. The resulting ranges for the ten input parameters $\boldsymbol{\xi}$ have thus been identified as: $[0.74, 0.76]$ for the Mach number $M = \xi_1$; $[0.55, 0.65]$ for the lift coefficient $C_L = \xi_2$; $[0.5, 2.0]$ for the four bending parameters ξ_3, ξ_4, ξ_5, and ξ_6 which are multiplied to the reference bending stiffness in each control cross-section (see Fig. 5); and $[0.02, 10.0]$ for the four torsion parameters ξ_7, ξ_8, ξ_9, and ξ_{10} which are multiplied to the reference torsion stiffness in each control cross-section (see Fig. 5).

2.3.2. *Latin Hypercube Sampling*

A Latin Hypercube Sampling (LHS) has been set on the parameter ranges determined manually and optimized for space filling. 100 samples have been generated to allow for a reasonable CPU time and to reach the empirical 10 samples per parameters. The minimum torsion value proved to be too aggressive and led to diverging wing shapes when associated with some other parameters. Thus, several sets of parameters did not lead to converged solutions (loss of 17 points). In the end, by combining the manual samples and the LHS samples, 101 points have been completely evaluated (about 10 points per parameter).

2.3.3. *Outputs*

The output quantities of interest are: the angle of attack α; the CFD code drag outputs (pressure-induced contribution $C_{D,\text{pressure}}$ and friction-induced contribution $C_{D,\text{friction}}$); the drag coefficients obtained from a far-field drag analysis (lift-induced contribution $C_{D,\text{lift induced}}$, wave contribution $C_{D,\text{wave}}$, and viscous contribution $C_{D,\text{viscous pressure}}$ which adds to the friction-induced contribution $C_{D,\text{friction}}$); the numerical spurious drag contribution ($C_{D,\text{spurious}}$); the CFD code pitching moment coefficient (C_m); and the wing shape characterized by the wing tip bend (U) and maximum twist (ϕ). The latter is defined as the relative differential vertical displacement between the leading and trailing edges aligned in the direction of the streamflow; see Hantrais-Gervois & Destarac (2015)

2.3.4. *Brief analysis of the results*

Fig. 7 displays various dependencies between the input parameters and the aerodynamic outputs. If some dependency can be drawn between the drag and the Mach number or lift coefficient, no simple explicit relationship can be derived with the bending and torsion parameters (not presented here). On the contrary, some dependencies can be simple: well-known linear dependence between lift induced drag and lift, spurious drag as a function of Mach number.[a] Some other outputs exhibit strong trends but depend on multiple parameters. Some cross dependencies between outputs are also displayed on Fig. 8. A high correlation can be denoted between wave drag and viscous pressure drag (top left plot). A less trivial near-correlation exists between the tip twist and the tip bend and other complex dependencies are also presented. This database contains many input parameters and outputs (aerodynamics and steady aeroelasticity).

Precise computations enable to extract well-known or subtle dependencies between the parameters for a flexible wing in transonic regime. These idiosyncrasies make it a highly valuable database for

[a]It should be noted that the strong dependence evidenced by the top right Fig. 7 is actually quadratic, and not linear.

uncertainty propagation and quantification.

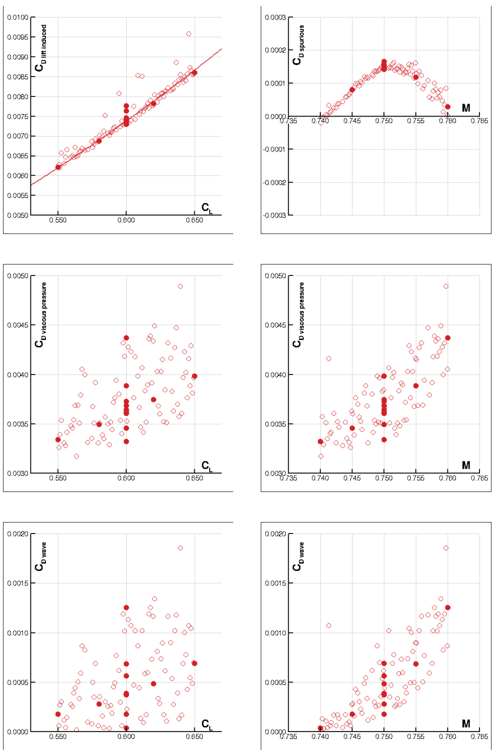

Fig. 7. Noticeable input-output dependencies for aerodynamics (● manual sampling set, ○ LHS set): $C_{D,\text{lift induced}}$ vs. C_L (top left); $C_{D,\text{spurious}}$ vs. M (top right); $C_{D,\text{viscous pressure}}$ vs. C_L (middle left); $C_{D,\text{viscous pressure}}$ vs. M (middle right); $C_{D,\text{wave}}$ vs. C_L (bottom left); $C_{D,\text{wave}}$ vs. M (bottom right).

3. Sparse polynomial surrogates

The computation of the marginal PDFs and first moments (mean, standard deviation, skewness, kurtosis...) of the output quantities of interest when the variability of the parameters $\boldsymbol{\xi}$ is accounted for, is done thank to surrogate models, or response surfaces. We more particularly focus on polynomial surrogates in this study.

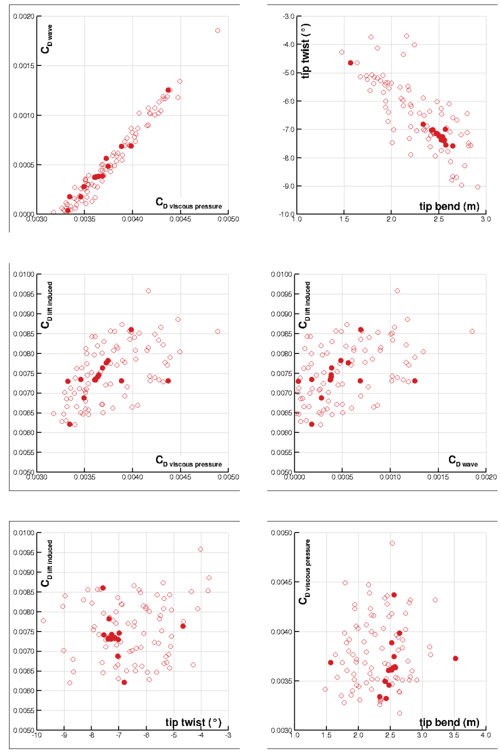

Fig. 8. Noticeable output-output dependencies (● manual sampling set, ○ LHS set): $C_{D,\text{wave}}$ vs. $C_{D,\text{viscous pressure}}$ (top left); ϕ (°) vs. U (m) (top right); $C_{D,\text{lift induced}}$ vs. $C_{D,\text{viscous pressure}}$ (middle left); $C_{D,\text{lift induced}}$ vs. $C_{D,\text{wave}}$ (middle right); $C_{D,\text{lift induced}}$ vs. ϕ (°) (bottom left); $C_{D,\text{viscous pressure}}$ vs U (m) (bottom right).

3.1. *Polynomial surrogates*

Polynomial surrogates for aerodynamic computations have been presented in *e.g.* Savin *et al*. (2016). We basically follow this presentation here. Let g be a generic physical/numerical model involving D parameters $\boldsymbol{\xi} = (\xi_1, \xi_2, \dots \xi_D) \in \mathcal{I} \subseteq \mathbb{R}^D$, such that the quantity of interest $y \in \mathcal{Y}$ is given as:

$$y = g(\boldsymbol{\xi})\,.$$

Let $\mathbb{P}^q[\boldsymbol{x}]$ be the set of D-dimensional polynomials with total order q with respect to $\boldsymbol{x} \in \mathbb{R}^D$. We first note that this set has cardinality $\#\mathbb{P}^q[\boldsymbol{x}] = P + 1$ such that:

$$P+1=\frac{(q+D)!}{q!D!}\,. \tag{1}$$

A polynomial surrogate model $\hat{g}_q$ of order q for the model $g \approx \hat{g}_q$ is obtained as:

$$\hat{g}_q = \arg\min_{\pi\in\mathbb{P}^q[x]} \frac{1}{2}\mathbb{E}\{|\pi(\Xi)-g(\Xi)|^2\}\,, \tag{2}$$

where $\mathbb{E}\{f(\Xi)\} := \int_{\mathbb{R}^D} f(\boldsymbol{x})\mathcal{P}_\Xi(\mathrm{d}\boldsymbol{x})$ is the mathematical expectation (average) of f, and $\Xi \sim \mathcal{P}_\Xi$ is the marginal PDF of the random parameters Ξ. The accuracy of this approximation may be assessed considering the limit of the mean-square norm $\mathbb{E}\{|\hat{g}_q(\Xi)-g(\Xi)|^2\}$ as $q \to +\infty$. However such a "convergence" does not necessarily holds, and it depends on the PDF $\mathcal{P}_\Xi$.

There are several ways to construct polynomial response surfaces and surrogates: embedded projection (this is the original spectral stochastic finite element method of Sun (1979) and Ghanem & Spanos (1991) which is highly intrusive), non-intrusive projection (*e.g.* collocation) (Ghiocel & Ghanem, 2002, Xiu & Hesthaven, 2005), or regression, whereby an ℓ_2 optimization problem is formed; see also Le Maître and Knio (2010) for a detailed introduction. For that purpose, a set of sampling points $\{\boldsymbol{\xi}_\ell\}_{\ell=1}^N$ where the model g has been evaluated, is needed in order to discretize the minimization problem (2). These observations are denoted by $\{y_\ell\}_{\ell=1}^N$ where $y_\ell = g(\boldsymbol{\xi}_\ell)$, $1 \leq \ell \leq N$.

3.2. *Polynomial chaos expansion*

From now on we assume that an orthonormal polynomial basis $\mathcal{B}$ of $L^2(\mathcal{I},\mathcal{P}_\Xi)$ is available. Then we construct the polynomial surrogate $\hat{g}_q$ of g by standard L^2 projection on the finite dimensional subspace of $L^2(\mathcal{I},\mathcal{P}_\Xi)$ spanned by the truncated family of orthonormal polynomials up to the total order q denoted by $\mathcal{B}^q = \{\psi_j\}_{j=0}^P$, where P is again given by Eq. (1). The orthonormalization of this basis reads:

$$\mathbb{E}\{\psi_j(\Xi)\psi_k(\Xi)\} = (\psi_j,\psi_k)_{L^2} = \delta_{jk}\,; \tag{3}$$

then:

$$\hat{g}_q(\boldsymbol{x}) = \sum_{j=0}^P g_j\psi_j(\boldsymbol{x}) = \sum_{j=0}^P (g,\psi_j)_{L^2}\psi_j(\boldsymbol{x})\,. \tag{4}$$

Such representations are referred to as polynomial chaos (PC) expansions in the dedicated literature, provided that the variable parameters Ξ follow a multidimensional Gaussian (normal) distribution $\mathcal{P}_\Xi = \bigotimes_{d=1}^D \mathcal{N}(0,1)$ (Ghanem & Spanos, 1991, Le Maître and Knio, 2010). They are otherwise called generalized polynomial chaos (gPC) expansions for other distributions (Soize & Ghanem, 2004, Xiu & Karniadakis, 2002).

The sampling points are typically used to evaluate the expansion coefficients $g_j = (g,\psi_j)_{L^2}$ in the foregoing representation. For example, they can be chosen as the N integration points of a quadrature rule $\boldsymbol{\Theta}^N = \{\omega_\ell,\boldsymbol{\xi}_\ell\}_{\ell=1}^N$, which provides with positive weights $\{\omega_\ell\}_{\ell=1}^N$ and nodes $\{\boldsymbol{\xi}_\ell\}_{\ell=1}^N$ in $\mathbb{R}^D$ such that for a smooth function $\boldsymbol{x} \mapsto f(\boldsymbol{x})$ one can evaluate its average by:

$$\mathbb{E}\{f(\Xi)\} \simeq \sum_{\ell=1}^N \omega_\ell f(\boldsymbol{\xi}_\ell)\,. \tag{5}$$

Using the quadrature rule above, the expansion coefficients are approximated by:

$$g_j \approx g_j^N = \sum_{\ell=1}^N \omega_\ell y_\ell \psi_j(\boldsymbol{\xi}_\ell)\,, \quad 0 \leq j \leq P\,.$$

This corresponds to the approximation $\hat{g}_q \approx \hat{g}_q^N = \sum_{j=0}^P g_j^N\psi_j$. Classical Gauss-Jacobi quadratures require q nodes to exactly integrate one-dimensional polynomials of order $2q-1$. The polynomial basis $\mathcal{B}^q$ adapted to the parameters PDF $\mathcal{P}_{\Xi}$ is constituted by the D-dimensional orthonormal polynomials $\psi_{\boldsymbol{j}}$, $\boldsymbol{j}=(j_1,j_2,\dots j_D)\in\mathbb{N}^D$, such that $\|\boldsymbol{j}\|_1=\sum_{d=1}^D j_d\leq q$ (note that they are $P+1$ such multi-indices according to Eq. (1) and it is understood here and below that they are simply reordered as single indices in the PC expansion (4)). These polynomials are given by:

$$\psi_{\boldsymbol{j}}(\boldsymbol{x})=\prod_{d=1}^D\psi_{j_d}(x_d)\,,\quad \|\boldsymbol{j}\|_1\leq q\,,$$

where $\{\psi_{j_d}\}_{j_d\geq 0}$ is the family of one-dimensional orthonormal polynomials with respect to the PDF of the d-th variable parameter. Therefore, $N\approx\lfloor\frac{q}{2}\rfloor^D$ sampling points are needed to exactly integrate D-dimensional polynomials of total order q. Using sparse quadrature rules (Gerstner & Griebel, 1998, Novak & Ritter, 1999, Smolyak, 1963) yields $N\approx 2^{\lfloor\frac{q}{2}\rfloor}P$ whenever $D\gg 1$, which may still be unaffordable for a complex model g: this is the so-called curse of dimensionality. Practical examples, though, reveal that the vector $\boldsymbol{g}=(g_0,g_1,\dots g_P)^{\mathsf{T}}\in\mathbb{R}^{P+1}$ of the expansion coefficients of $\hat{g}_q$ can have many negligible components (Savin *et al.*, 2016), so that it is *compressible* in the terminology of the theory of compressed sensing, or compressive sampling (CS) (Candès *et al.*, 2006, Candès & Romberg, 2007, Candès & Wakin, 2008, Donoho, 2006). In other words, its sparsity:

$$S=\#\{j;\,|g_j|>\delta\}\tag{6}$$

for some tolerance $0\leq\delta\ll\mathbb{E}\{|\hat{g}_q(\Xi)|\}$, say, is such that $S\ll P$. In this setting it is argued that only a number of samples N proportional to the compressed size S, rather than the uncompressed size P, of the sought signal (in the present case the surrogate model $\hat{g}_q$) is needed in order to reconstruct it. The reconstruction of the expansion coefficients from a sparsity argument is outlined in the next section. There we basically follow the introductory paper of Candès & Wakin (2008) on the underlying CS theory.

3.3. *Sparse reconstruction by ℓ_1–minimization*

Evaluating the gPC expansion (4) at the sampling points $\boldsymbol{\Theta}^N=\{\boldsymbol{\xi}_\ell\}_{\ell=1}^N$ we arrive at the $N\times(P+1)$ system:

$$\boldsymbol{y}=\boldsymbol{M}\boldsymbol{g}+\boldsymbol{\epsilon}\,,\tag{7}$$

where $\boldsymbol{y}=(y_1,y_2,\dots y_N)^{\mathsf{T}}\in\mathcal{Y}^N$ is the vector of observations, and $\boldsymbol{M}$ is the so-called $N\times(P+1)$ measurement matrix given by $[\boldsymbol{M}]_{\ell j}=\psi_j(\boldsymbol{\xi}_\ell)$. Since the gPC expansion truncated at the total order q is not complete for the exact representation of g, a truncation error vector $\boldsymbol{\epsilon}$ has been accounted for in the foregoing process. Eq. (7) is ill-posed and has infinitely many solutions whenever $N\ll P$, unless additional constraints of the sought solution $\boldsymbol{g}$ are accounted for. Regularized versions of Eq. (7) exist for this case, which in turn ensure its well-posedness. Together with a sparsity constraint, this can be accommodated by reformulating Eq. (7) as the following ℓ_0–minimization problem where a sparsest approximation $\boldsymbol{g}^\star$ of $\boldsymbol{g}$ is sought for as:

$$\boldsymbol{g}^\star=\arg\min_{\boldsymbol{h}\in\mathbb{R}^{P+1}}\{\|\boldsymbol{h}\|_0;\;\|\boldsymbol{M}\boldsymbol{h}-\boldsymbol{y}\|_2\leq\varepsilon\}\,,\tag{$P_{0,\varepsilon}$}$$

for some tolerance $0\leq\varepsilon\ll 1$ on the polynomial chaos truncation; the norms above are defined by $\|\boldsymbol{h}\|_m=(\sum_{j=0}^P|h_j|^m)^{\frac{1}{m}}$ for $m>0$ and $\|\boldsymbol{h}\|_0=\#\{j;\,h_j\neq 0\}$ otherwise. The solution of the problem $(P_{0,0})$ is not always unique and the latter is also NP-hard to solve, that is, the time needed to solve it is exponential with respect to P (as for example the brute force algorithm for optimization). However it can be relaxed by convex ℓ_1–minimization, known as Basis Pursuit Denoising (BPDN) (Chen *et al.*, 1998):

$$g^\star = \arg\min_{h\in\mathbb{R}^{P+1}} \{\|h\|_1;\ \|Mh-y\|_2 \le \varepsilon\}\,. \tag{$P_{1,\varepsilon}$}$$

For sufficiently sparse coefficients g and some conditions on the measurement matrix M, $(P_{0,0})$ and $(P_{1,0})$ share the same unique solution. Consequently the strategy for the present study is to solve $(P_{1,\varepsilon})$ with N runs of the aeroelastic model g significantly lower than the number of coefficients to be identified. Indeed CS theory shows that it is relevant provided that the target solution g is actually sparse, or nearly sparse (compressive), and some constraints on the measurement matrix are fulfilled.

One of such requirements for the successful recovery of a sparse vector by $(P_{1,\varepsilon})$ is small mutual coherence of the columns of the measurement matrix, or their near orthogonality. The *coherence* $0<\mu(\Theta^N,\mathcal{B}^q)\le 1$:

$$\mu(\Theta^N,\mathcal{B}^q) = \max_{\substack{1\le j,k\le P+1\\ j\neq k}} \frac{|m_j^\mathsf{T} m_k|}{\|m_j\|_2\|m_k\|_2}\,,$$

where $m_j = (\psi_j(\xi_1),\psi_j(\xi_2),\dots\psi_j(\xi_N))^\mathsf{T}$ stands for the j–th column of M, is a measure of how close to orthogonality the measurement matrix is. Based on this coherency measure, the following theorem from Candès & Romberg (2007) asserts that if $\hat{g}_q$ is sufficiently sparse in $\mathcal{B}^q$, the recovery of its gPC coefficients by ℓ_1–minimization is exact.

Theorem 3.1. *Assume that $\hat{g}_q$ is S–sparse on the gPC basis $\mathcal{B}^q$, that is g has at most S non-zero entries. Then if N sampling points $\Theta^N = \{\xi_\ell\}_{\ell=1}^N$ are selected at random to form the measurement matrix M, and:*

$$N \ge C\cdot\mu(\Theta^N,\mathcal{B}^q)\cdot S\cdot\log P \tag{8}$$

for some constant $C>0$, the solution of $(P_{1,0})$ is exact with "overwhelming" (sic) probability.

More precise results with structured random measurement matrices are given by *e.g.* Rauhut & Ward (2012). It should be noted that the role of coherence in this result is transparent. The smaller the coherence is, the closer the measurement matrix is to a unitary matrix, and the fewer sampling points are needed. This result also states that $(P_{1,0})$ can be effectively solved without any prior knowledge of the number of non-zero coefficients, their orders (locations), or their amplitudes. In other words, the reconstruction procedure permitted by the CS theory is *non adapted* because it identifies the sparsity pattern of a signal in its sparsifying basis, and the leading components at the same time. This procedure can therefore efficiently capture the relevant information of a sparse signal without trying to comprehend that signal (Candès & Wakin, 2008) contrary to the proposal of Blatman & Sudret (2010). This is clearly a much desirable feature for practical industrial applications.

The previous Theorem 3.1 is however not entirely satisfactory from a practical point of view because (i) it does not allow for some truncation error, or noisy/imprecise observations; (ii) it does not deal with approximately sparse vectors, for which a large subset of entries are negligible rather than strictly zero. These shortcomings may be alleviated simultaneously as established by Candès *et al.* (2006). To achieve this, a constraint on the measurement matrix M needs be added to gain robustness in CS, the so-called *restricted isometry property* (RIP) also quoted as the uniform uncertainty principle. For each integer $S\in\mathbb{N}^*$, the isometry constant δ_S of M is defined as the smallest number such that:

$$(1-\delta_S)\|h_S\|_2^2 \le \|Mh_S\|_2^2 \le (1+\delta_S)\|h_S\|_2^2$$

for all S-sparse vectors $h_S\in\mathcal{Y}_S := \{h\in\mathbb{R}^{P+1};\ \|h\|_0\le S\}$. Then M is said to satisfy the RIP of order S if, say, δ_S is not too close to 1. This property amounts to saying that all S–column submatrices of $\boldsymbol{M}$ are numerically well-conditioned, or S columns $\boldsymbol{m}_{j_1}, \boldsymbol{m}_{j_2} \dots \boldsymbol{m}_{j_S}$ selected arbitrarily in $\boldsymbol{M}$ are nearly orthogonal. The following theorem by Candès *et al.* (2006), Candès & Wakin (2008) then states that $(P_{1,\varepsilon})$ can be solved efficiently:

Theorem 3.2. *Assume $\delta_{2S} < \sqrt{2} - 1$. Then the solution $\boldsymbol{g}^\star$ to $(P_{1,\varepsilon})$ satisfies:*

$$\|\boldsymbol{g}^\star - \boldsymbol{g}\|_2 \leq C_0 \frac{\|\boldsymbol{g}_S - \boldsymbol{g}\|_1}{\sqrt{S}} + C_1 \varepsilon$$

for some $C_0, C_1 > 0$. Here $\boldsymbol{g}_S$ is $\boldsymbol{g}$ with all but the S largest entries set to zero.

This result calls for several comments. First, it is more general than Theorem 3.1 since, if the signal is exactly S–sparse, $\boldsymbol{g} = \boldsymbol{g}_S$ and the reconstruction is exact whenever $\varepsilon = 0$ (noiseless case). Second, it deals with all signals, not the S–sparse ones solely. Third, it is deterministic and does not involve any probability. Lastly, the bound $\sqrt{2} - 1$ on δ_{2S} is the one originally proposed by Candès & Wakin (2008) but it can be improved as proposed by *e.g.* Mo & Li (2011); such improvements are an active field of research at present.

3.4. *Uncertainty Quantification (UQ)*

Once the polynomial surrogate model $\hat{g}_q$ has been derived, a mean output functional of the quantity of interest y can be estimated by:

$$\mathbb{E}\{f(y)\} \simeq \int_{\mathcal{I}} f(\hat{g}_q(\boldsymbol{x}))\mathcal{P}_{\Xi}(\mathrm{d}\boldsymbol{x}),$$

where $y \mapsto f(y)$ is a regular function on $\mathcal{Y}$. The mean μ is obtained for $f(y) = y$, the variance σ^2 is obtained for $f(y) = (y - \mu)^2$, the skewness γ_1 for $f(y) = (\frac{y-\mu}{\sigma})^3$, the kurtosis β_2 for $f(y) = (\frac{y-\mu}{\sigma})^4$, *etc.* Owing to the orthonormality (3) of the polynomials of $\mathcal{B}^q$ the mean is simply $\mu = g_0$ and the variance is $\sigma^2 = \sum_{j=1}^P g_j^2$. Higher-order moments may be computed with the formulas derived in Savin & Faverjon (2017) for the continuous orthogonal polynomials of the Askey scheme.

Sensitivity indices may be computed alike. Denoting by $\mathcal{I}_d$ the set of indices corresponding to the polynomials of $\mathcal{B}^q$ depending only on the d–th variable parameter ξ_d, the main-effect gPC-based Sobol' indices are given by (see for example Sudret (2008)):

$$\mathcal{S}_d = \frac{1}{\sigma^2} \sum_{j \in \mathcal{I}_d} g_j^2 , \tag{9}$$

invoking again the normalization condition (3). More generally, if $\mathcal{I}_{d_1 d_2 \dots d_s}$ is the set of indices corresponding to the polynomials of $\mathcal{B}^q$ depending only on the parameters $\xi_{d_1}, \xi_{d_2}, \dots \xi_{d_s}$, the s–fold joint sensitivity indices are:

$$\mathcal{S}_{d_1 d_2 \dots d_s} = \frac{1}{\sigma^2} \sum_{j \in \mathcal{I}_{d_1 d_2 \dots d_s}} g_j^2 .$$

In the following application to the aeroelastic database, we will primarily consider the main-effect sensitivity indices $\mathcal{S}_d$.

4. Application to ALBATROS database

We now apply the foregoing procedure to the non-adaptive computation of the gPC coefficients $\boldsymbol{g}$ of the surrogate models $\hat{g}_q$ for the output quantities of interest of Sect. 2.3.3. We use the $N = 101$ sampling points drawn manually and by LHS as explained in Sect. 2.3, so that the chaos basis $\mathcal{B}^q$ is constituted by Legendre polynomials because the latter are orthogonal with respect to the uniform measure. Note in passing that the uniform probability measure is the one arising from Jaynes' maximum entropy principle (Jaynes, 1957a,b) when the sole imposed constraint is the compactness of its support. We consider a total order $q = 3$, thus $P = 285$ since $D = 10$ here, with $\xi_1 = M$, $\xi_2 = C_L$, ξ_3 through ξ_6 for the bending parameters, and ξ_7 through ξ_{10} for the torsion parameters. The coherence for the present sampling set and representation basis is $\mu(\mathbf{\Theta}^{101}, \mathcal{B}^3) \simeq 0.94$ and the sparsity of the polynomial surrogates is observed to be $S \simeq 10$. Thus Eq. (8) yields $N \gtrsim 50$. A common observation is that $N \gtrsim 4S \simeq 40$ is usually enough for a successful recovery.

We subsequently apply BPDN $(P_{1,\varepsilon})$ to compute $\boldsymbol{g}$. For that purpose we use the Spectral Projected Gradient Algorithm (SPGL1) developed by van den Berg & Friedlander (2008) and implemented in the Matlab package `SPGL1` (van den Berg & Friedlander, 2007) to solve this ℓ_1-minimization problem. The tolerance was fixed at $\varepsilon = 10^{-5}$ and we were able to find a solution for all surrogates with this *a priori* choice without resorting to cross-validation, for example. Further investigations should be carried on on this topic, though. It should also be noted that no particular sampling strategy, such as stratification, low-discrepancy series, or preconditioning, has been applied at this stage to select the sampling set. Alternative strategies are outlined in several recent works; see for example Doostan & Owhadi (2011), Hampton & Doostan (2015), Mathelin & Gallivan (2012), Peng *et al*. (2014), Rauhut & Ward (2012), Schiavazzi *et al*. (2014).

The root mean-square errors of the output quantities of interest C_m (pitching moment coefficient), $C_{Ds} = C_{D,\text{pressure}} + C_{D,\text{friction}}$ (skin drag coefficient), $C_{Dv} = C_{D,\text{lift induced}} + C_{D,\text{wave}} + C_{D,\text{viscous pressure}} + C_{D,\text{friction}}$ (far-field drag coefficient), α (angle of attack), U (wing tip bend), and ϕ (wing tip twist) are gathered in the Table 1. The root mean-square error e_2 is defined by (note that it is directly related to the chosen tolerance ε):

$$e_2 = \sqrt{\frac{\sum_{\ell=1}^{N} |y_\ell - \hat{g}_q(\boldsymbol{\xi}_\ell)|^2}{\sum_{\ell=1}^{N} |y_\ell|^2}}\,.$$

The mean μ and standard deviation σ are gathered in the Table 1 as well, while the main-effect sensitivity indices $\mathcal{S}_1$ through $\mathcal{S}_{10}$ of Eq. (9) are gathered in Table 2. The strong dependence of the wing tip bend U on the second bending parameter ξ_4 can be observed from these results ($\mathcal{S}_4 = 62\%$). The noticeable dependence of the wing tip twist ϕ on the third bending parameter ξ_5 may also be observed ($\mathcal{S}_5 = 32\%$) . The PDFs are displayed on Fig. 9. They were estimated from $N_s = 100,000$ interrogations of the gPC surrogates $\hat{g}_q$ and smoothing out the resulting histograms by a normal kernel density function (Wand & Jones, 1995). They are compared with a standard Gaussian distribution with the same mean and standard deviation. These comparisons are quantified by the Kullback-Leibler divergence $D_{\text{KL}}(\mathcal{P}||\mathcal{Q})$ defined by Kullback & Leibler (1951):

$$D_{\text{KL}}(\mathcal{P}||\mathcal{Q}) = \int_{\mathcal{Y}} \log \frac{\mathcal{P}(y)}{\mathcal{Q}(y)} \mathcal{P}(\mathrm{d}y)\,,$$

where $\mathcal{P}$ is the reconstructed marginal PDF of the output quantity of interest y, and $\mathcal{Q} \equiv \mathcal{N}(\mu, \sigma)$ is the standard Gaussian distribution it is compared to. It is computed in the Table 1 below for the various output quantities of interest considered here.

We have also quantified the dependence of the lift-induced drag coefficient $C_{D,\text{lift induced}}$ and spurious drag coefficient $C_{D,\text{spurious}} = C_{Ds} - C_{Dv}$ on the various parameters. The tolerance for the computation of their gPC coefficients by $(P_{1,\varepsilon})$ was fixed at $\varepsilon = 10^{-6}$ and $\varepsilon = 10^{-9}$, respectively. The root mean-

Table 1. Root mean-square error, mean, standard deviation, and Kullback-Leibler divergence from a Gaussian distribution of the output quantities of interest computed by ℓ_1–minimization with $N = 101$ sampling points.

	C_m	C_{Ds}	C_{Dv}	α (°)	U (m)	ϕ (°)
μ	-11.63e-02	219.83e-04	218.69e-04	2.53	2.16	-6.10
σ	1.55e-02	6.53e-04	6.28e-04	0.20	0.26	0.80
e_2	4.70e-03	0.45e-03	0.35e-03	1.88e-03	1.65e-03	3.97e-03
D_{KL}	1.10e-02	1.11e-02	1.39e-02	1.40e-02	0.50e-02	0.62e-02

Table 2. Main-effect sensitivity indices of the output quantities of interest computed by ℓ_1–minimization with $N = 101$ sampling points.

	C_m	C_{Ds}	C_{Dv}	α	U	ϕ
$\mathcal{S}_1$	0.048e-01	2.657e-01	2.286e-01	0.003e-01	0.002e-01	0.105e-01
$\mathcal{S}_2$	0.036e-01	1.841e-01	1.682e-01	1.333e-01	0.007e-01	0.011e-01
$\mathcal{S}_3$	0.052e-01	0.136e-01	0.136e-01	0.005e-01	0.974e-01	0.021e-01
$\mathcal{S}_4$	0.229e-01	0.000e-01	0.002e-01	0.154e-01	6.187e-01	0.461e-01
$\mathcal{S}_5$	0.046e-01	0.000e-01	0.002e-01	0.001e-01	1.137e-01	3.186e-01
$\mathcal{S}_6$	0.280e-01	0.038e-01	0.027e-01	0.010e-01	0.016e-01	0.128e-01
$\mathcal{S}_7$	0.078e-01	0.002e-01	0.004e-01	0.840e-01	0.196e-01	0.774e-01
$\mathcal{S}_8$	0.005e-01	0.001e-01	0.001e-01	0.722e-01	0.109e-01	0.506e-01
$\mathcal{S}_9$	0.322e-01	0.115e-01	0.149e-01	0.003e-01	0.010e-01	0.034e-01
$\mathcal{S}_{10}$	0.285e-01	0.073e-01	0.191e-01	0.390e-01	0.057e-01	0.411e-01

square error, mean, standard deviation, and Kullback-Leibler divergence from a Gaussian distribution are gathered in the Table 3, while the main-effect sensitivity indices are gathered in Table 4. Their PDFs are displayed on Fig. 10, using again $N_s = 100,000$ evaluations of the gPC surrogates and smoothing with a normal kernel density function. These results confirm the strong dependence of $C_{D,\text{spurious}}$ on the Mach number ξ_1 (for $\mathcal{S}_1 = 94\%$), and to a lesser extent that of $C_{D,\text{lift induced}}$ on the lift coefficient ξ_2 (for $\mathcal{S}_2 = 48\%$).

Table 3. Root mean-square error, mean, standard deviation, and Kullback-Leibler divergence from a Gaussian distribution of the lift-induced and spurious drag coefficients computed by ℓ_1–minimization with $N = 101$ sampling points.

	$C_{D,\text{lift induced}}$	$C_{D,\text{spurious}}$
μ	72.60e-04	0.88e-04
σ	3.52e-04	0.46e-04
e_2	0.74e-03	14.19e-03
D_{KL}	1.16e-02	15.18e-02

5. Conclusions

In this paper we have applied a non intrusive technique to reconstruct surrogates (response surfaces) for the uncertainty quantification and robust optimization of aeroelastic computations, taking the example of the ALBATROS configuration. This technique is a regression approach, whereby the surrogates are polynomial chaos expansions in the variable parameters space. It typically uses structured sampling set, *e.g.* Gauss quadratures, to compute the expansion coefficients in this basis. However we have shown in this study that invoking a "sparsity-of-effects" feature, the framework of compressed sensing is applicable to the present case. Accordingly, an ℓ_1–minimization algorithm has been implemented to reconstruct polynomial

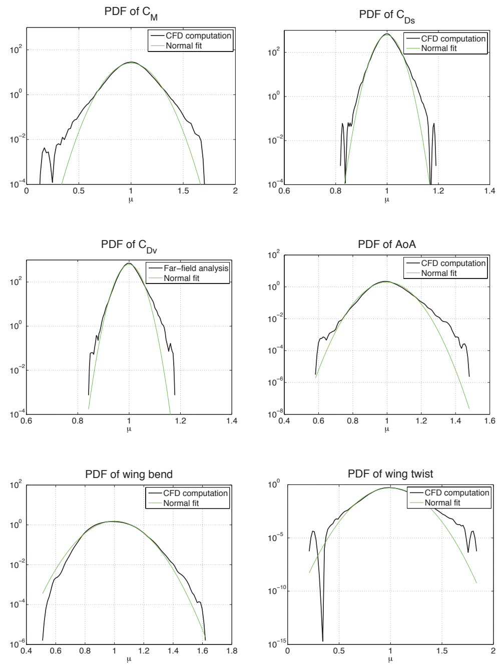

Fig. 9. PDFs of the pitching moment (top left), skin drag (top middle), and far-field drag (top right) coefficients, and angle of attack (bottom left), wing tip bend (bottom middle), and wing tip twist (bottom right) computed by ℓ_1–minimization with $N = 101$ sampling points. Comparison with a standard Gaussian distribution (green curve) with the same mean and standard deviation.

Table 4. Main-effect sensitivity indices of the output quantities of interest computed by ℓ_1–minimization with $N = 101$ sampling points.

	$C_{D,\text{lift induced}}$	$C_{D,\text{spurious}}$
$\mathcal{S}_1$	0.004e-01	9.417e-01
$\mathcal{S}_2$	4.811e-01	0.005e-01
$\mathcal{S}_3$	0.014e-01	0.001e-01
$\mathcal{S}_4$	0.007e-01	0.021e-01
$\mathcal{S}_5$	0.000e-01	0.002e-01
$\mathcal{S}_6$	0.082e-01	0.056e-01
$\mathcal{S}_7$	0.071e-01	0.013e-01
$\mathcal{S}_8$	0.020e-01	0.012e-01
$\mathcal{S}_9$	0.096e-01	0.010e-01
$\mathcal{S}_{10}$	0.044e-01	0.008e-01

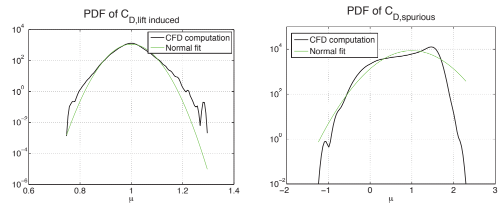

Fig. 10. PDFs of the lift-induced (left) and spurious (right) drag coefficients computed by ℓ_1–minimization with $N = 101$ sampling points. Comparison with a standard Gaussian distribution (green curve) with the same mean and standard deviation.

surrogates in a regression approach. In addition, it requires unstructured sampling sets orders of magnitude smaller than the ones required by the widespread sampling rules (Shannon's sampling theorem or Gauss quadratures, for example), and typically smaller than the dimension of the polynomial space where the surrogates are sought for. The procedure is also non-adaptive in the sense that it identifies both the amplitude of the leading expansion coefficients and their order in the polynomial basis. At last, its application to the aeroelastic database of the present study has clearly demonstrated its ability to handle a parameters space of high dimension–thus alleviating to some extent the "curse of dimensionality." Therefore, the sparse reconstruction technique outlined here constitutes a promising direction for future developments in large-scale industrial applications involving complex geometries and flows.

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